Assignment 1

Deployment Environment and Cloud Platform Recommendation

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GitHub Commit:

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AI Use Statement: I used ChatGPT to brainstorm deployment options, explore ideas for the deployment diagram, improve grammar and sentence clarity, and simplify technical concepts related to cloud services and AI deployment. I reviewed the suggestions, checked relevant information against course materials and official documentation, and made the final decisions about the deployment approach and recommendations.

Section 1: Use case and readiness snapshot

Use case- Card fraud scoring for a credit union

1.1 Use Case Description

My use case is card fraud scoring for a credit union. The users of this system are fraud analysts, fraud investigators, transaction processing staff, risk and compliance teams and the IT team to manage the system. Customers are also affected by it because their cards will be checked when making a transaction. This system helps the credit union decide whether the transaction should be approved, declined or sent to a fraud employee for review. It uses 3 years of training data like number of transactions (160 million), amount, merchant category, time, location, whether the card's physically present and the customer's usual spending habits. Historical legitimate or fraudulent transactions can also be used to train the model. The type of model in this case would be a supervised binary classification model for example a gradient boosted decision tree to produce a fraud risk score for each transaction. The lifecycle stage that matters most is monitoring and retraining because criminals change tactics and accuracy reduces withing weeks (Huyen, 2022).

1.2 Why This Is Production AI

It must be a production AI because it would be used on real card transactions and needs high uptime It must handle many transactions at the same time. A wrong prediction may lead to consequences like allowing fraudulent transaction or a false decline at checkout, stranding a member. Neither a notebook or a prototype can promise 50ms responses, failover or rollbacks.

1.3 Readiness Assessment

Refer to Appendix A

1.4 Lowest Readiness Scores and Deployment Impact

The two lowest scoring dimensions are Risk and Organizational. Risk is lowest because our system handles sensitive financial information and the credit union must ensure that data is protected and system is tested to reduce incorrect decisions before launching it. Organization is also a concern because someone needs to be responsible for monitoring the model, checking its performance and retraining. If these responsibilities are unclear, there would be problems after deployment.

Section 2: Deployment environment evaluation

2.1 Refer to Appendix B: Table B1

My recommendation: Hybrid

2.2 Refer to Appendix B: Table B2

2.3 I recommend a hybrid environment for the credit union's card-fraud scoring system. I chose this option because the system needs to protect sensitive customer information and check transactions quickly. The two most important framework questions are data sensitivity (1) and latency(1). Keeping the main transaction-processing system on-premises gives the credit union more control over customer data and helps reduce delays when checking card payments. The workloads would be divided between the two environments. On-premises infrastructure would handle real-time transaction authorization and critical fraud checks, while AWS would be used for model training, approved data storage, model testing, and performance monitoring. This allows the credit union to use AWS resources without moving all transaction processing to the cloud. The scorecard rule, with an average of 2.5 out of 5 agrees with this

recommendation, hence there is no recommendation to justify. Therefore, I would choose hybrid deployment because it balances speed and data control with the benefits of cloud computing.

2.4 Refer to Appendix C: C2
I compared the hybrid environment with on premises. On cloud is excluded by the hard limits. I estimated the costs over three years using assumptions about transaction volume, computing resources, storage, services, and staff. Based on these estimates, the hybrid option would cost about US$1.3 million, compared with approximately US$2.0 million for the fully on-premises option.

2.5 **Tipping Point**

I would change my recommendation from hybrid to all-cloud if the 99th-percentile round-trip time between the card processor and AWS Canada (Central) stayed below 15 ms for a full month, and the auditor approved cloud scoring using tokenized transaction data. Until both conditions are met, live fraud scoring will remain on-premises to meet the 50 ms response-time limit and maintain data control.

Section 3: Cloud platform and service selection

3.1 Platform- Amazon Web Services(AWS).

Why AWS.
1. Amazon SageMaker AI covers training, pipelines, model registry and drift monitoring in one managed service, which suits a team with no ML-operations staff (Amazon Web Services, n.d.-c).
2. AWS Direct Connect gives a private, dedicated link to our data centre, with a VPN as backup (Amazon Web Services, n.d.-d).
3. Data residency: Canada (Central) in Montreal keeps data in Canada.
4. AWS has a broad range of cloud services and partners, which could make it easier to find support and people with relevant skills.

Why not other platforms?
i) Microsoft Azure would also be a good option, especially if the credit union already relied heavily on Microsoft tools, but that existing dependency is not assumed here.
ii) Google Cloud may be less suitable if the team has less experience with its services and ecosystem.

3.2 Refer to Appendix C: Table C1: AWS Service Map

3.3 I would use containers for live fraud scoring because transactions come in continuously, responses must be quick, and the model needs to run on the credit union's own systems. For weekly training, I would use managed containers in Amazon SageMaker AI because training takes longer but only happens occasionally. For pipeline triggers and alerts, I would choose serverless services like AWS Lambda because these tasks run only when needed. This approach keeps the main fraud-scoring system reliable while reducing the work needed to manage supporting tasks.

3.4 **Build vs. Buy**

Criterion	**Managed ML Platform (Amazon SageMaker AI)**	**Self-Managed Stack (Own GPUs and Open-Source Tools)**
Time to value	Faster because AWS manages the infrastructure.	Longer to setup and maintain.

Control and customization	Allows a custom model and features but follows AWS platform requirements.	Provides full control over the model, tools, and infrastructure.
Cost over time	Pay for services, estimated at much lower cost.	Requires hardware and an additional employee, expensive.
Required expertise	Requires ML skills.	Requires additional MLOps and on-call expertise.
Decision	**Chosen:** Faster to deploy and requires less infrastructure maintenance.	Not chosen because it requires more staff, setup time, and ongoing support.

3.5

i) **Plan:** I would first identify the system's goals the required response time, the need for data privacy and the expected costs. I would also decide which tasks stay on-premises, and which tasks move to AWS.
ii) **Ready:** Before building anything, I would set up AWS accounts or environments for development and production. I would configure IAM permissions enable encryption and logging and create budget alerts to monitor spending.
iii) **Adopt:** I would test the cloud model in shadow mode for four weeks before letting it handle 5% of live traffic. I would compare the cloud model results with existing fraud rules. I would use infrastructure as code and versioned model images to allow a rollback if any cloud problems occur.
iv) **Govern & Operate:** I would review cloud model drift every week. I would adjust fraud thresholds each month with the fraud team, quarterly access reviews and disaster recovery testing of cloud disconnection (Beyer et al., 2016).

3.6 Areas where I would be most locked in include SageMaker AI pipelines and Model Monitor that require proprietary tools from AWS, and Direct Connect agreement, which might be difficult to change. To minimize the chances of lock-in, I will use open-source tools like Boost, maintain training code in Git and save models in formats that are transportable. This will help me to retrain the model on a different platform whenever necessary, while the on-premises scoring container remains independent from AWS.

Section 4: Risks and decision

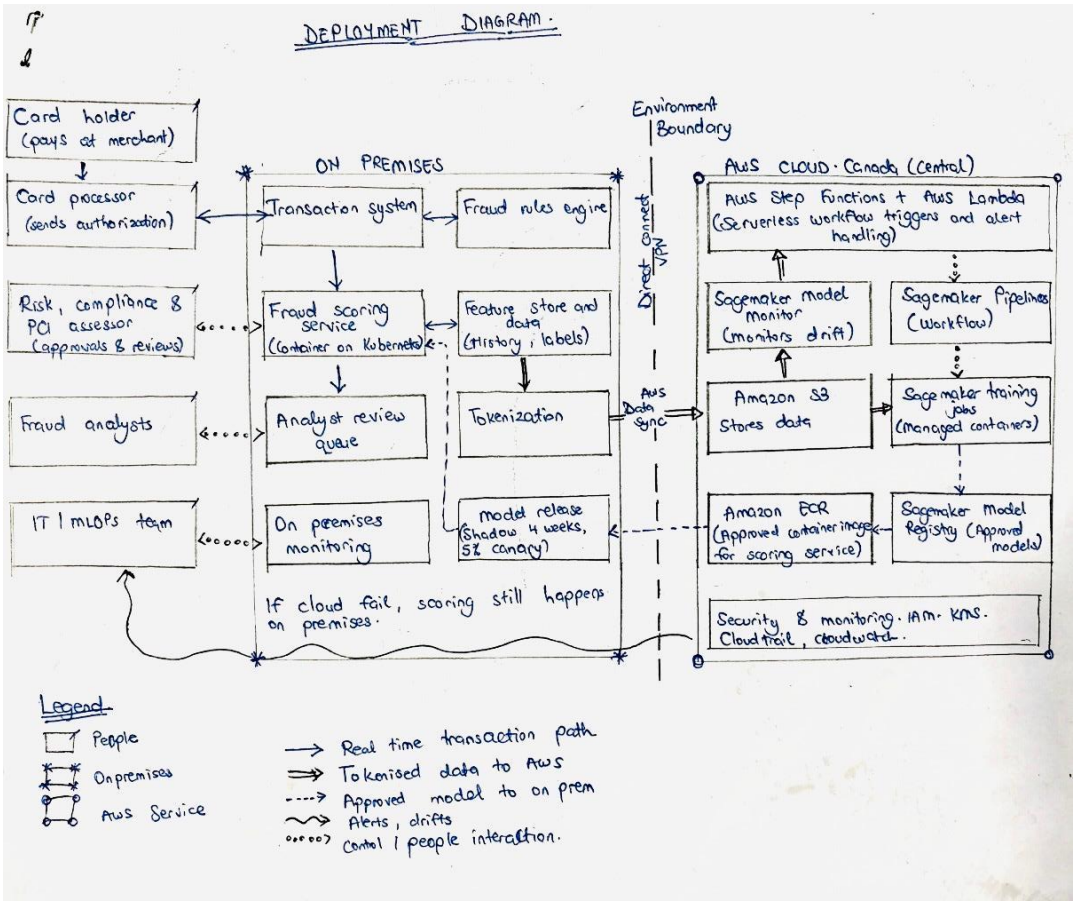
4.1

Risk	**Likelihood/ Impact**	**Mitigation**
Technical: Fraud patterns may change making the model miss new patterns.	High/High	Review performance and any drift, retrain the model if needed, collect analyst feedback and keep previous version of the model and rules for feedback.
Security: Failure to tokenize the data correctly may expose the card details, for example in 2019 Capital one's cloud misconfiguration breach exposed over 100 million customer records (Omar, 2026).	Medium/ High	Tokenize sensitive card details before sending data to AWS, also restrict access and review security controls with the PCI assessor every three months.

Cloud/ network outages: Connection issues/ a cloud outage could disrupt services, for example the AWS outage in 2021 showed how it can affect businesses at scale (AWS, 2021).	Low/ High	Keep live fraud scoring on premises so payments can continue if AWS becomes unavailable. Use Direct Connect with a VPN backup and keep practicing the recovery process every three months.

4.2 I would delay the launch until the main risks are fixed. Before making a Go (Ship) decision the on-premises scoring service needs to finish four weeks of shadow testing. During this time the percentile latency must stay under 50 ms at three times peak traffic. The testing also needs to show least 20% more fraud detected without any increase in false declines. A security and PCI review must confirm that raw card numbers are not stored in AWS. A cloud-link failure test must show that payments can still go through. Formal model-risk approval and on-call support must be ready. Infrastructure and risk readiness scores must both be least 3, out of 5. If all these conditions are met I would start with a 5% canary rollout.

Section 5: Communication and professional practice



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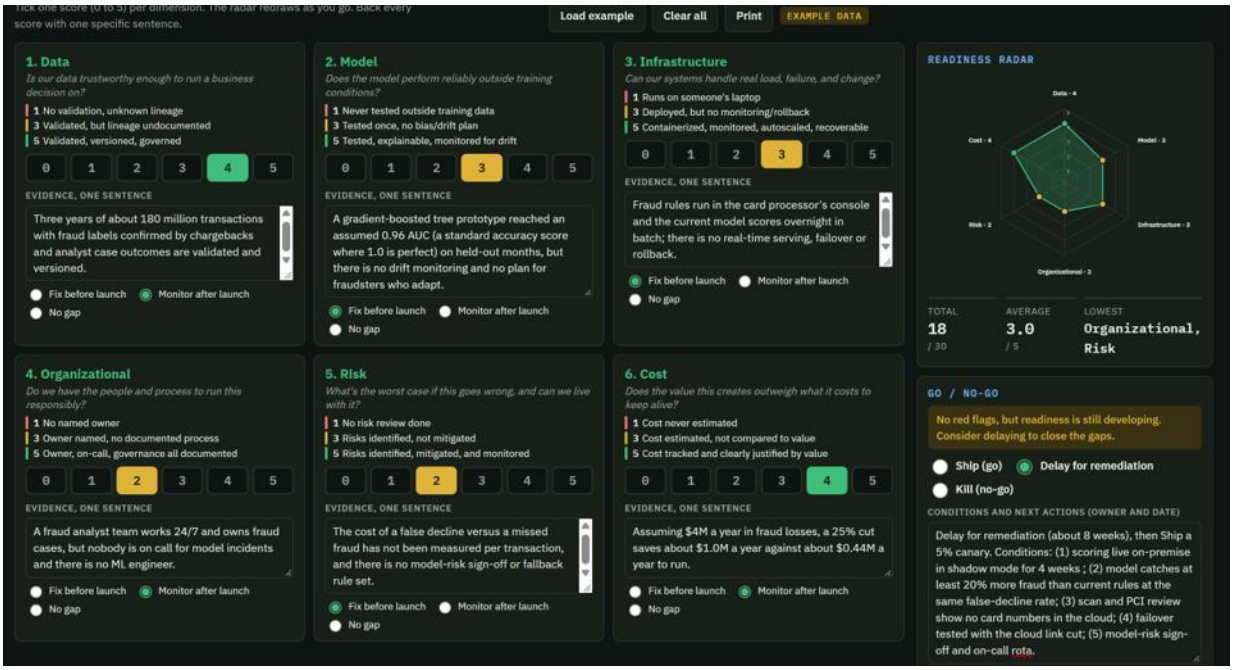
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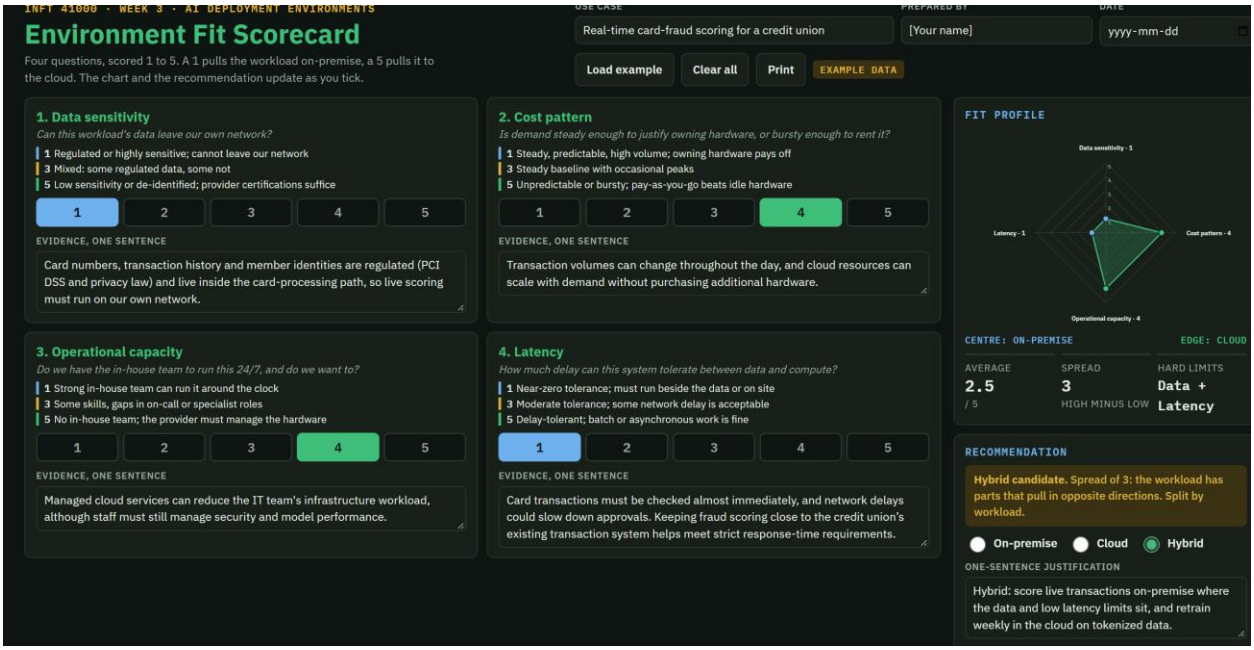
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Appendix A: Completed Readiness Scorecard



Readiness	Score	Evidence	Effect on Deployment
Data	4/5	The credit union has historical transaction data and confirmed fraud cases available for training.	Monitor after launch and continue improving the quality and timeliness of fraud labels.
Model	3/5	A gradient-boosted tree prototype reached 0.96 AUC on held-out months, but there is no drift monitoring.	Monitor model performance after launch and retrain when performance decreases.
Infrastructure	3/5	The credit union already has transaction-processing infrastructure, but the system needs reliable model serving, monitoring, and integration.	Fix before launch so that it can handle transaction volume.
Organizational	2/5	The credit union may have fraud and IT expertise, but clear responsibility for monitoring, retraining, and maintaining the model needs to be established.	Fix before launch by assigning ownership for the AI system and its ongoing maintenance.
Risk	2/5	The system handles sensitive financial information and influence transaction decisions, so security, privacy and compliance controls must be confirmed.	Fix before launch because security and compliance issues need to be sorted.
Cost	4/5	Scalable cloud resources control costs compared with adding all infrastructure internally.	Monitor after launch and track cloud usage to stay within the planned budget.

Appendix B: Table B1: Environment Fit Scorecard and comparison templates



Question	Facts about my use case	Score
1. Data sensitivity	The system handles sensitive financial information so the credit union needs control over data access and processing.	1
2. Cost pattern	I notice that transaction volumes can change throughout the day and cloud resources can scale with demand without purchasing hardware.	4
3. Operational capacity	Managed cloud services can reduce the IT team's infrastructure workload, although staff must still manage security and model performance.	4
4. Latency	Keeping fraud scoring near the credit union's transaction system helps fraud scoring meet strict response-time requirements.	1

Average: ___2.5___ Spread (highest minus lowest): ___3___ Data or latency scored 1? Yes / No
Rule says: a 1 on data or latency rules out public cloud alone. A spread of 3 or more means hybrid. Otherwise 2.4 or less is on-premise, 3.6 or more is cloud, and in between operational capacity breaks the tie.

Table B2. Environment comparison for my use case

Dimension	On-premise	Cloud	Hybrid
Data control	Full control; card numbers never leave, but all of the payment-card security standard	Approved data is processed on AWS, requiring strong security controls.	Raw card data stays; only tokenized data (card numbers swapped for random stand-ins) goes to

	(PCI DSS) applies to us		the cloud.
Cost model	The credit union pays for its own servers, upgrades, and maintenance.	AWS charges for the resources used, but costs may rise during busy periods.	The credit union pays for existing systems and additional AWS services.
Scalability	More transactions may require upgrades to existing servers.	AWS resources increase when transaction volumes rise.	Existing systems take care of transactions while AWS helps with growing AI work.
Operational burden	The IT team keeps maintaining servers, updates models, and fixes problems.	AWS manages infrastructure. The credit union watches model performance and security.	The IT team manages both environments, which adds some complexity.
Best fit for	Partly; serving only	Suitable for model training and experiments that need extra computing power.	Best overall because it keeps critical transactions in-house and uses AWS for AI training and monitoring.

Appendix C: Service map and cost templates

Table C1. Service map (one platform: AWS)

Building Block	**Named Service**	**Service Model (IaaS/PaaS/SaaS)**	**Container / Serverless**	**Why**
Training compute	Amazon SageMaker AI Training Jobs	PaaS	Managed training job	Trains the model on tokenized historical data without requiring the credit union to manage its own training servers.
Serving / hosting	On-premises containerized scoring service running on the credit union's own Kubernetes infrastructure; Amazon Elastic Container Registry (Amazon ECR) stores approved images.	On premises: own infrastructure; Amazon ECR: PaaS	Container	Keeps live fraud scoring close to the transaction system to meet the fast response-time requirement, even if AWS is unavailable.
Data storage	Amazon Simple Storage Service (Amazon S3) for tokenized	PaaS	Not applicable	Provides durable storage for approved data and model

	training data and model artifacts			files, with encryption and access controls.
Managed AI / ML	Amazon SageMaker AI Pipelines, Model Registry, and Model Monitor	PaaS	Managed services	Automates model training workflows, tracks approved model versions, and helps identify performance changes or data drift.
Orchestration	AWS Step Functions and AWS Lambda	PaaS	Serverless	Starts workflows when new data arrives and triggers alerts without requiring dedicated servers.
Data movement	AWS DataSync	PaaS	Managed service	Transfers approved tokenized data from the credit union's on-premises environment to Amazon S3.
Security and monitoring	AWS Identity and Access Management (IAM), AWS Key Management Service (KMS), AWS CloudTrail, Amazon CloudWatch, and AWS Budgets	PaaS	Managed services	Controls access, manages encryption keys, records activity, monitors system health, and alerts the team to unexpected spending.

Table C2. Three-year cost comparison (order of magnitude)

Cost line	**Assumptions and sources**	**Option A: Hybrid**	**Option B: All on-premises**	**Uncertainty**
Compute	165,000 transactions/day (3x peak) is a modest load: three scoring servers (two active, one DR) at about $40,000 = $120,000, plus 15% a year for support, power and colocation = $54,000. A: weekly training of about 4 hours on one ml.m5.4xlarge-class CPU instance at about $0.922 an hour (about $250 a year; tree models need no GPU) plus experiments and notebooks at about $2,000 a month = $72,000. B: own training server about $150,000 plus 15% a year = $68,000. Amazon Web Services. (n.d.). Getting started with AWS Pricing Calculator. AWS Documentation. https://docs.aws.amazon.com/pricing-calculator/latest/userguide/getting-started.html 2026/10/08	$0.25M	$0.36M	Medium

Storage and data transfer	A: 5 TB in S3- $110/ month, DataSync- $8/ month Direct Connect 1 Gbps port and transfer about $230 a month = $ 2760 plus on-premise storage and feature store $30,000. B: on-premise storage and feature store $80,000. Amazon Web Services. (n.d.). Getting started with AWS Pricing Calculator. AWS Documentation. https://docs.aws.amazon.com/pricing-calculator/latest/userguide/getting-started.html 2026/10/08	$0.034M	$0.08M	Medium
Managed services and licences	A: SageMaker AI Pipelines and Model Monitor and related services about $1,500 a month = $54,000, plus on-premise Kubernetes, monitoring and rules tooling about $20,000 a year = $60,000. B: MLOps and monitoring tooling about $50,000 a year. Amazon Web Services. (n.d.). Amazon SageMaker AI pricing. https://aws.amazon.com/sagemaker/ai/pricing/	$0.12M	$0.15M	Medium
Staff time to operate	Hybrid assumes two full-time staff members; on-premises assumes three. Each costs an estimated $150,000 per year, including employment costs.	$0.90M	$1.35M	High
Three-year total	Rounded total of the cost categories above.	About $1.3M	About $2.0M	Uncertain: staff time